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# Support Vector Machine (SVM)
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SVM stands for Support Vector Machine. SVM is a supervised machine learning algorithm. It is mainly used for classification problems. SVM can also be used for regression, which is called Support Vector Regression (SVR). The main objective of SVM is to find the best decision boundary between different classes. SVM selects the decision boundary that provides the maximum margin between the classes. The data points that are closest to the decision boundary are called Support Vectors. Support vectors are the most important data points because they determine the position of the decision boundary.

2. Hyperplane

The decision boundary in SVM is called a Hyperplane. A hyperplane separates different classes in the feature space. In a 2-dimensional space, the hyperplane is a line. In a 3-dimensional space, the hyperplane is a plane. In higher-dimensional spaces, it is called a hyperplane.

3. Margin

Margin is the distance between the decision boundary and the nearest data points of each class. SVM tries to maximize the margin. The nearest data points to the decision boundary are called support vectors. A larger margin generally helps the model achieve better generalization. Therefore, SVM is also known as a maximum-margin classifier.

4. Support Vectors

Support vectors are the training data points that are closest to the decision boundary. They determine the position and orientation of the optimal hyperplane. Support vectors are critical to the SVM solution. If important support vectors change, the decision boundary can also change. Other training points that are far away from the boundary generally have less direct influence on the final boundary. Therefore, SVM gets its name from these support vectors.

5. Hard Margin SVM

Hard-margin SVM assumes that the classes are perfectly linearly separable. It does not allow any training point to be misclassified. It also does not allow points to violate the required margin. Hard-margin SVM tries to find the hyperplane with the maximum possible margin. Hard-margin SVM is sensitive to noise and outliers. Therefore, hard-margin SVM is generally not suitable for noisy real-world datasets.

6. Soft Margin SVM

Soft-margin SVM is used when the data is not perfectly separable. It allows some training points to violate the margin. It can also allow some points to be misclassified. SVM introduces a slack variable to allow these violations. The parameter C controls how strongly these violations are penalized. Soft-margin SVM is more suitable for real-world datasets.

7. C Parameter

C is an important hyperparameter of SVM. C controls the penalty for classification errors and margin violations. A high C gives a high penalty for misclassification. With high C, SVM tries harder to classify training points correctly. This can result in a smaller/narrower margin and a more complex boundary. A very high C can increase the risk of overfitting. A low C allows more margin violations. Low C generally produces a wider margin and stronger regularization. A very low C can lead to underfitting.

8. Kernel Trick

Real-world data is not always linearly separable. A straight-line decision boundary may not be sufficient for nonlinear data. SVM uses the kernel trick to handle nonlinear relationships. The kernel trick allows SVM to work as if the data were mapped into a higher-dimensional feature space. This mapping is performed implicitly rather than explicitly constructing all higher-dimensional features. In the transformed space, a linear hyperplane can potentially separate the classes. Therefore, kernels allow SVM to create nonlinear decision boundaries in the original feature space.

9. Linear Kernel

Linear kernel is used when the data is approximately linearly separable. It creates a linear decision boundary. It is generally faster than nonlinear kernel SVM. It is particularly useful for high-dimensional data. Linear SVM is commonly used in text classification and other sparse high-dimensional problems.

10. Polynomial Kernel

Polynomial kernel is used to model nonlinear relationships. It creates a polynomial-based decision boundary. The complexity of the polynomial kernel depends partly on the degree parameter. A higher degree can create a more complex decision boundary. Excessively high degree can increase the risk of overfitting.

11. RBF Kernel

RBF stands for Radial Basis Function. RBF is one of the most commonly used nonlinear kernels in SVM. It can create complex nonlinear decision boundaries. RBF depends strongly on the gamma (γ) parameter. Gamma controls how strongly an individual training point influences the surrounding region. A low gamma produces a smoother decision boundary. A high gamma produces a more complex and localized decision boundary. Therefore, very high gamma can cause overfitting.

12. Gamma

Gamma is mainly important for kernels such as RBF. Gamma determines the influence of individual training samples. Low gamma means a training point has a wider area of influence. This generally produces a smoother decision boundary. High gamma means a training point has a smaller and more localized area of influence. This can produce a highly complex decision boundary. Very high gamma can cause overfitting. Very low gamma can cause underfitting.

13. Hinge Loss

SVM classification is commonly associated with hinge loss. Hinge loss penalizes predictions that are incorrect or insufficiently far from the decision boundary.

If a correctly classified point is sufficiently outside the margin, its hinge loss is zero. If a point violates the margin, it receives a positive loss. Therefore, hinge loss helps SVM focus on correctly classified points that are close to or inside the margin.